

Exp No:2 Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

Aim:

To upload and analyze the data set given in csv format and perform data preprocessing and visualization.

Description:

Use sample data set sales-data.csv.

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

file_path = 'sales-data-new.csv'
df = pd.read_csv(file_path)
print(df.head())
print(df.isnull().sum())
df['Sales'] = df['Sales'].fillna(df['Sales'].mean())
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
print(df.describe())
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()
print(product_summary)
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
```

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plt.title('Total Sales by Product')

plt.show()

df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)

sales_over_time = df.groupby('Date').agg({

    'Sales': 'sum'

}).reset_index()

plt.figure(figsize=(10, 6))

plt.plot(sales_over_time['Date'], sales_over_time['Sales'])

plt.xlabel('Date')

plt.ylabel('Total Sales')

plt.title('Sales Over Time')

plt.xticks(rotation=45)

plt.show()

pivot_table = df.pivot_table(

    values='Sales',

    index='Region',

    columns='Product',

    aggfunc='sum',

    fill_value=0

)

print(pivot_table)

correlation_matrix = df[['Sales', 'Quantity']].corr()

print(correlation_matrix)

plt.figure(figsize=(8, 6))

sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')

plt.title('Correlation Matrix')

plt.show()
```

output:

	Date	Product	Sales	Quantity	Region
0	01-01-2023	Product A New	255.0	14	North
1	02-01-2023	Product B New	197.5	13	South
2	03-01-2023	Product A New	278.0	15	North
3	04-01-2023	Product C New	370.0	16	East
4	05-01-2023	Product B New	232.0	14	West

Date 0

Product 0

Sales 0

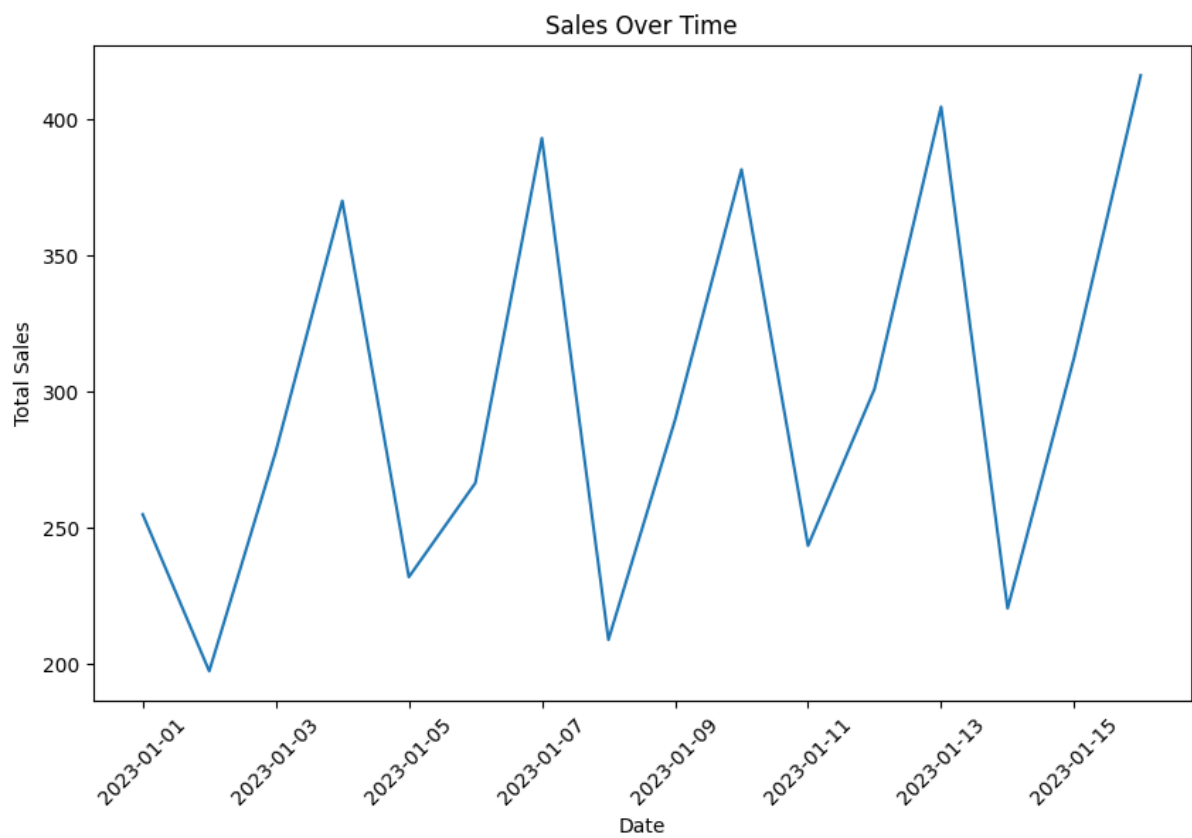
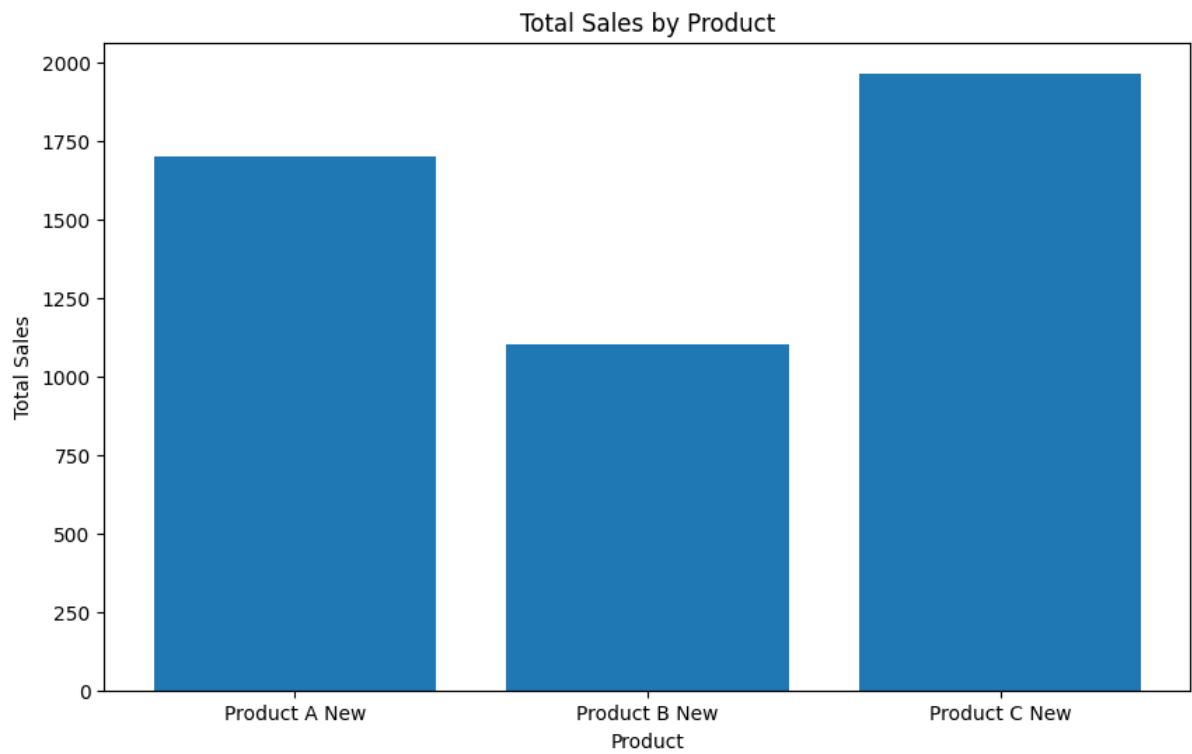
Quantity 0

Region 0

dtype: int64

	Sales	Quantity
count	16.000000	16.000000
mean	298.125000	15.375000
std	73.635929	1.746425
min	197.500000	13.000000
25%	240.625000	14.000000
50%	283.750000	15.500000
75%	372.875000	17.000000
max	416.000000	18.000000

	Product	Sales	Quantity
0	Product A New	1702.5	93
1	Product B New	1102.5	67
2	Product C New	1965.0	86



Product Product A New Product B New Product C New

Region

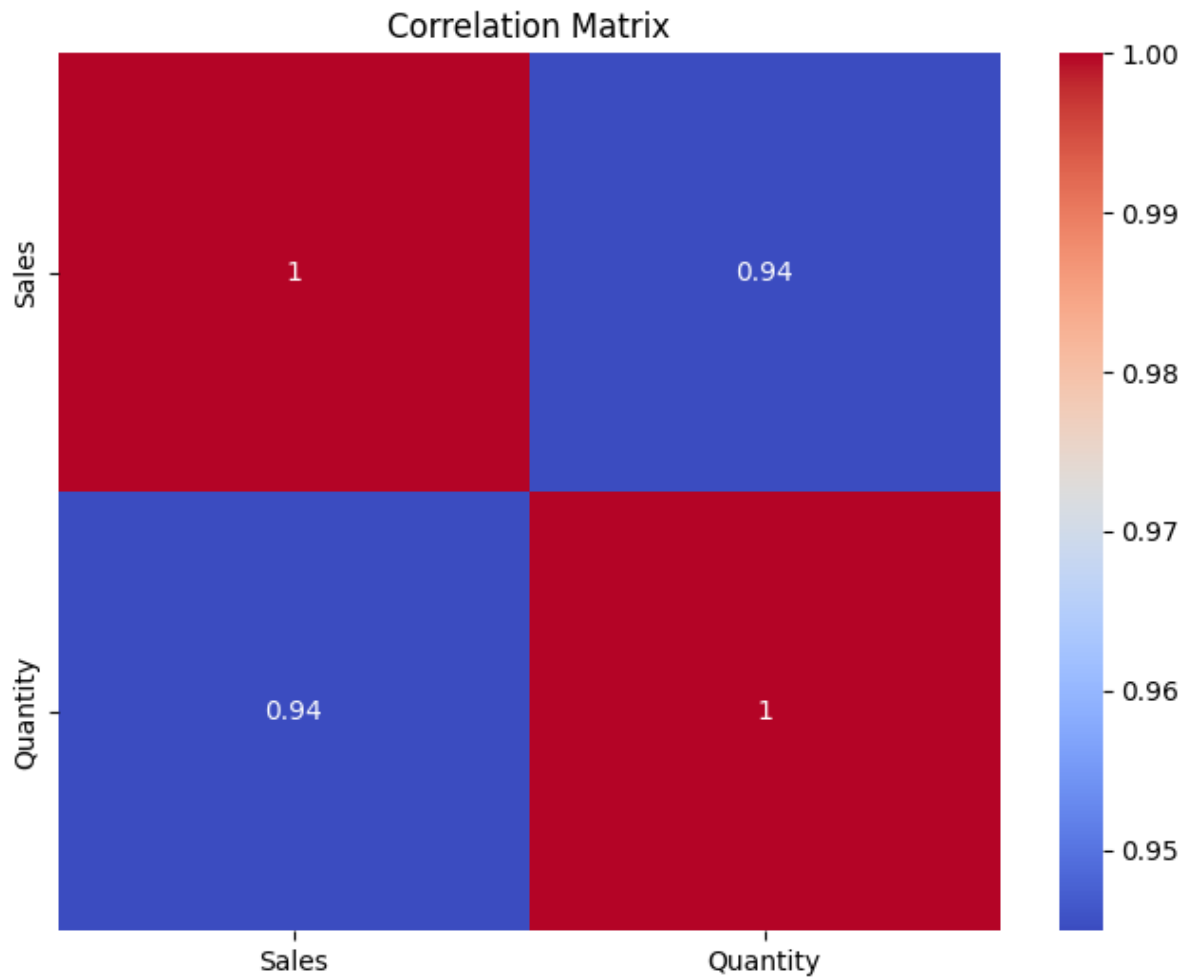
East 0.0 0.0 1965.0

North	1702.5	0.0	0.0
South	0.0	627.0	0.0
West	0.0	475.5	0.0

Sales Quantity

Sales	1.000000	0.944922
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Quantity	0.944922	1.000000
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Result:

The dataset was successfully loaded and analysed using pandas. The preprocessing operation was performed successfully, and no important missing values were present.